Final paper and project guidelines:

1. Purpose of the Capstone Paper

The capstone paper documents the student's independent research or applied project. The purpose of the paper is to demonstrate the ability to:

- Identify and formulate a relevant research or applied problem
- Review and interpret relevant scientific literature
- Apply appropriate data science or engineering methods
- Present results and validate findings
- Communicate technical work clearly in a scientific format

The capstone paper is the **primary artifact used to evaluate the capstone project**.
All papers must be prepared using the **IEEE conference proceedings format** and submitted in **PDF format with Latex and/or Overleaf** generated from **LaTeX or Overleaf**.

2. Paper Format

Students must use the **IEEE conference template**:

- IEEE conference proceedings format (https://tinyurl.com/y5rw3926)
- The ***IEEEtran class*** file is used to format your paper and style the text. All margins, column widths, line spaces, and text fonts are prescribed; please do not alter them.

All requirements for the IEEE conference paper, such as font size, margins, figure sizes, needs to be preserved.

Typical length of the final capstone paper is:
6–10 pages (IEEE format), excluding references and appendices.

3. Required Structure of the Paper

Although minor variations may exist depending on the project, the capstone paper should normally include the following sections.

3.1. Title and Author Information

Include:

- Title of the project
- Student name
- Affiliation (American University of Armenia)
- Supervisor name
- Anyother contributor

3.2. Abstract

The abstract should summarize the project in **150–250 words**, including:

- The problem addressed
- The methods used
- The key results
- The main contribution of the work

The abstract should allow readers to quickly understand the purpose and outcomes of the project.

3.3. Introduction

The introduction must clearly explain:

- The **problem being addressed**
- Why the problem is **important**
- The **research gap** in existing work
- The **objectives and research questions**

The introduction should follow the **“funnel approach”**, beginning with a broad context and narrowing to the specific problem addressed by the project.

3.4. Literature Review

This section should summarize and analyze **relevant scientific literature**, including:

- Key existing approaches
- Strengths and limitations of previous work
- How the current project relates to existing research

The literature review must demonstrate that the student:

- Understands the existing body of work
- Can critically evaluate prior research
- Can position their project within the field.

This section typically ranges from **1 to 4 pages**.

3.5. Methodology

The methodology section explains **how the problem is addressed**.
It should describe:

- Data sources or datasets used
- Algorithms, models, or analytical techniques
- System architecture or implementation design
- Experimental setup
- Tools, software, or frameworks used

Students must also justify **why the chosen method was selected over alternatives**.

3.6. Results

This section presents the **findings of the project**.
Typical elements include:

- Model performance metrics
- Experimental outcomes
- Figures, tables, and visualizations
- Outputs of implemented systems or algorithms

Results must be **clearly presented and supported by evidence**.

3.7. Analysis and Validation

Students must analyze and validate their results.
This section should include:

- Interpretation of results
- Comparison with existing work or baseline methods
- Validation techniques (experiments, benchmarks, simulations, etc.)
- Discussion of limitations

A key question to address is: Did the project meet its objectives?

3.8. Discussion and Conclusion
This section should summarize:

- Main contributions of the project
- Key findings
- Implications of the results
- Limitations of the work
- Possible future improvements or research directions

The conclusion must clearly state **how the project objectives were achieved**.

3.9. References
All references must follow **IEEE citation style**.
Requirements:

- Every citation in the text must appear in the reference list.
- References should include **recent and relevant scientific sources**, such as:
    - Journal articles
    - Conference papers
    - Books
    - Technical reports

3.10. Appendices (Optional)
Appendices may include:

- Additional experiments
- Detailed derivations
- Extended tables or figures
- Implementation details

4. Submission

Students must submit the following items separately through Moodle:

4.1. Capstone Paper (PDF)
The final paper must be submitted as a **PDF file** generated from the IEEE conference template.
The PDF must correspond exactly to the submitted source files.

4.2. Project Files
Students must submit a **compressed archive (.zip)** containing all project materials necessary to reproduce the results.
The archive must include:

4.2.2. Code
All code used for the project, including:

- Data preprocessing scripts
- Analysis scripts
- Model training code
- Visualization code
- Simulation or experimental scripts

The code should be organized and executable.
Examples of acceptable formats:

- Python (.py)
- R (.R, .rmd, .qmd)
- Jupyter notebooks (.ipynb)

4.2.3. Data

All datasets used in the project must be included.
If datasets cannot be shared:

- provide a **download link**
- provide **instructions for obtaining the data**

Students must clearly document:

- data sources
- preprocessing steps
- variable descriptions

4.2.4. LaTeX / Overleaf Source Files

Students must also submit the **complete LaTeX source of the paper**, including:

- main.tex
- bibliography files (.bib)
- figures
- additional .tex files if used

These files must allow the paper to be compiled without errors.

5. Project Structure

Students are strongly encouraged to organize the submission archive using the following structure (example):

```
capstone_project/
│
├── paper/
│   ├── main.tex
│   ├── references.bib
│   ├── figures/
│   └── paper.pdf
│
├── code/
│   ├── data_preprocessing.py
│   ├── analysis.py
│   ├── models/
│   └── visualization/
│
├── data/
│   ├── raw_data/
│   └── processed_data/
│
└── README.md
```

6. Documentation

Each project must include a **README file** explaining:

- project objective
- required software and libraries
- steps required to reproduce results
- how to run the code
- how figures and tables in the paper were generated

7. Environment and Dependencies

Students must specify the computational environment used in the project.
Examples include:

- Python version
- R version
- required packages and libraries

Recommended methods:

- requirements.txt (Python)
- environment.yml (conda)
- renv.lock (R)

8. Reproducibility Requirement

A project is considered reproducible if:

- the code runs without modification
- the same results presented in the paper can be generated
- figures and tables can be recreated from the provided scripts

Projects that cannot be reproduced may receive **penalties in evaluation**.

8.1. One-Command Reproduction Rule

The project must allow the main results of the paper to be reproduced using **a single command**.

8.2. Figures Must Be Generated by Code

All figures in the paper must be **generated programmatically** from the data.
Students should **not manually edit figures**.

9. Academic Integrity

Students must ensure that:

- all code submitted is their own work or properly cited
- external libraries are clearly referenced
- data sources are properly acknowledged